

LAB REPORT

EXPERIMENT 4 : MEASURING SKELETAL MUSCLE OXYGEN CONSUMPTION WITH NIRS



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ABSTRACT

Near-infrared spectroscopy (NIRS) was used to measure hemodynamic and metabolic responses in forearm skeletal muscle during venous and arterial occlusions at two levels of static-hold exercise intensity. A PortaMon continuous-wave NIRS (CW-NIRS) device recorded changes in oxyhemoglobin (HbO), deoxyhemoglobin (HbR), and total hemoglobin (HbT) concentrations in two subjects. Heart rate was extracted from resting NIRS signals using fast Fourier transform (FFT), yielding 52 bpm for Subject 1 and 72 bpm for Subject 2. Forearm blood flow was estimated from the linear slope of HbT during venous occlusion. Blood flow was found to increase with exercise intensity for Subject 2 but not for Subject 1, with HbT slopes ranging from 0.071 to 0.154 $\mu\text{M/s}$ (Subject 2) and 0.690 to 0.154 $\mu\text{M/s}$ (Subject 1) at low and high intensities, respectively. Tissue oxygen consumption (VO_2) was estimated from the HbO slope during arterial occlusion, yielding values of 0.025 to 0.78 $\mu\text{M/s}$ (Subject 2) and 0.146 to 0.185 $\mu\text{M/s}$ (Subject 1), suggesting that Subject 1 could have higher forearm-muscle mass than Subject 2. Slope uncertainties were estimated from the covariance matrix of the linear fits, yielding uncertainties of $\pm 0.003\text{--}0.010$ $\mu\text{M/s}$ for venous occlusion and $\pm 0.0003\text{--}0.0031$ $\mu\text{M/s}$ for arterial occlusion. Higher exercise intensity produced greater oxygen consumption in both subjects, consistent with increased metabolic demand. These results demonstrate the utility of CW-NIRS combined with vascular occlusion for noninvasive assessment of skeletal muscle vascular and metabolic function.

INTRODUCTION¹

The purpose of this experiment was to use near-infrared spectroscopy (NIRS) to investigate hemodynamic and metabolic changes in forearm muscle during venous and arterial occlusions. Measurements of oxyhemoglobin (HbO) and deoxyhemoglobin (HbR) concentration changes were used to estimate blood flow and tissue oxygen consumption.

Near-infrared spectroscopy (NIRS) is an effective way to image biological systems in a while minimizing invasiveness. NIRS is used to observe changes in hemoglobin concentrations through the measurement of light intensities that go through the tissue in question. This technique depends on the **Modified Beer-Lambert Law** (MBLL) as follows:

$$OD_{\lambda} = \log\left(\frac{I_o}{I}\right) = \epsilon_{\lambda} \cdot c \cdot L$$

OD_{λ} is the optical density of the tissue with respect to the wavelength λ ; I_o is the incident light intensity; I is the transmitted light intensity; ϵ_{λ} is the extinction coefficient of the chromophore ($\text{mM}^{-1}\text{cm}^{-1}$), c is the concentration in mM of the chromophore; and L is the distance in cm between the light input and the output points. Due to scattering effects, infrared light does not travel in a straight line through human tissues; it follows a banana-like path. Because of this, the NIRS source and detector must be placed on the same side of the tissue, roughly 2 to 4 cm apart¹.

Most NIRS devices monitor concentration changes of two chromophores: oxygen-hemoglobin (HbO) and deoxygen-hemoglobin (HbR). To get a good estimate of these two variables, two wavelengths from the optical window will be selected for the light source, as seen in Figure 3. Applying the MBLL gives one equation for each wavelength. To solve for HbO and HbR

concentration changes, the following 2 by 2 system is needed:

$$\begin{bmatrix} \Delta HbR \\ \Delta HbO \end{bmatrix} = \frac{1}{L} \begin{bmatrix} \epsilon_{HbR, \lambda_1} & \epsilon_{HbO, \lambda_1} \\ \epsilon_{HbR, \lambda_2} & \epsilon_{HbO, \lambda_2} \end{bmatrix}^{-1} \begin{bmatrix} \Delta OD(\Delta t, \lambda_1) / DPF(\lambda_1) \\ OD(\Delta t, \lambda_2) / DPF(\lambda_2) \end{bmatrix},$$

Figure 1: Modified Beer-Lambert Law (MBLL)¹

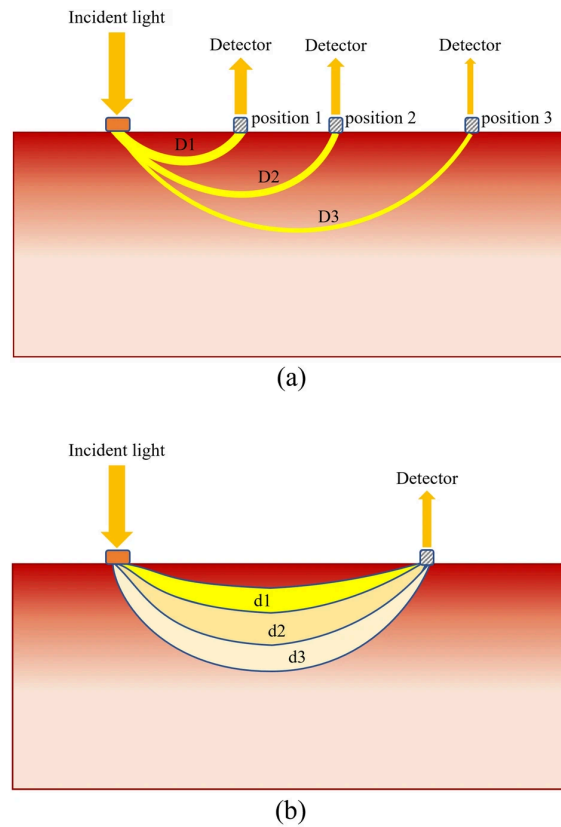


Figure 2: Near-infrared spectroscopy (NIRS) light scattering diagram and source-detector positioning².

Where $DPF(\lambda)$ is the differential path length factor (DPF) that is used to correct the path length from source-detector distance to the estimated banana-shape length.

Changes in hemoglobin concentration will result in changes in light intensity detected by the detector. By measuring the light intensity and then solving two MBL equations with two unknowns, HbO and HbR concentration changes can be calculated. This type of modelling is called continuous-wave NIRS (CW-NIRS), which is limited to the estimation of relative changes of hemoglobin since it does not model scattering. More advanced techniques such as frequency domain NIRS (FD-NIRS) and time domain NIRS (TD-NIRS) do estimate the absolute concentration change.

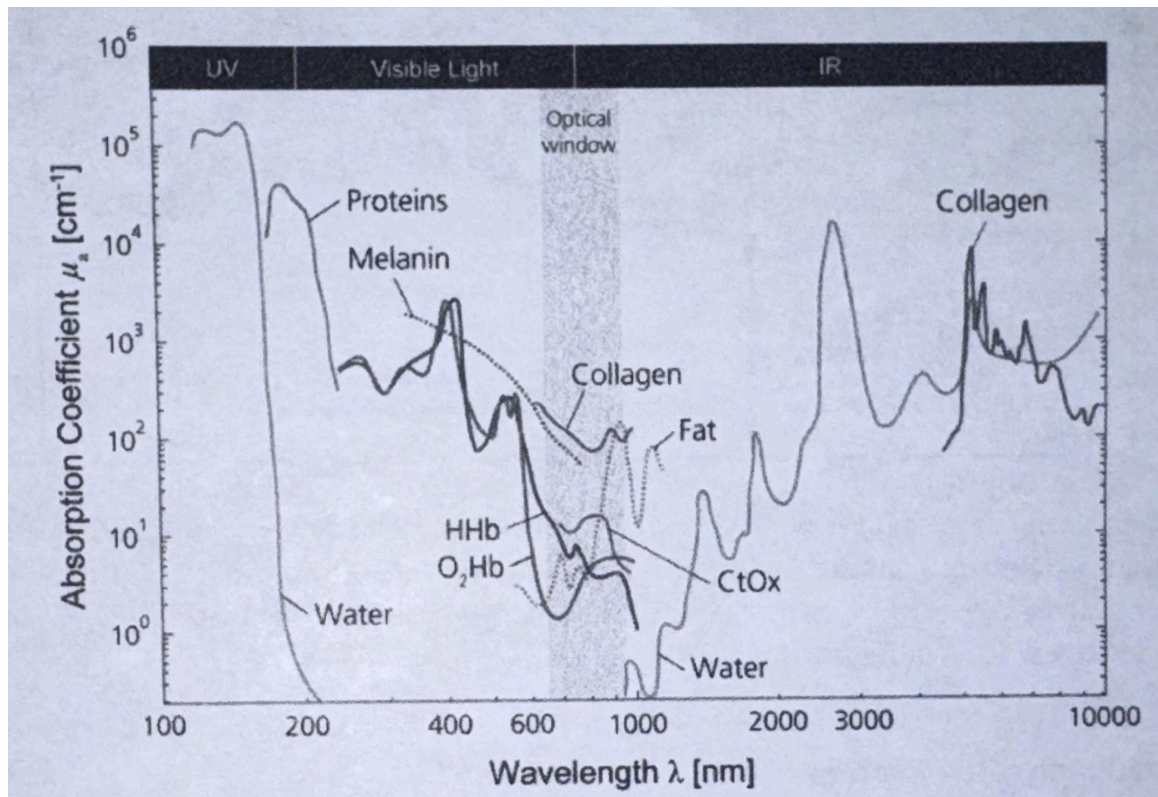


Figure 3¹: Absorption spectra of oxyhemoglobin (HbO) and deoxyhemoglobin (HbR) in the near-infrared range. The highlighted “optical window” (approximately 700–900 nm) shows the wavelengths where light penetration is optimized, allowing for clear measurements of HbO and HbR using near-infrared spectroscopy.

Venous Occlusion of the Forearm

Combining NIRS and venous occlusion allows for a measure of the blood flow and oxygen consumption. Venous occlusion blocks the return of blood while leaving arterial inflow unchanged. To do so, the pressure of the cuff is rapidly increased to a value above venous pressure but below arterial pressure, typically around 50 mmHg¹. Under these conditions, blood continues to enter the forearm through the arteries but cannot leave through the veins. As a result, blood accumulates within the tissue, producing an increase in total hemoglobin concentration ($HbT = HbO + HbR$). Since the rate of increase in total hemoglobin is proportional to the amount of blood entering the tissue, the initial slope of the HbT curve can be used to estimate forearm blood flow. During the short occlusion period, oxygen extraction by the tissue continues, causing HbO to decrease and HbR to increase slightly. However, the dominant effect is the accumulation of blood volume inside the forearm.

Blood Flow Measurement From Venous Occlusion

Forearm blood flow is determined from the rate of change of total hemoglobin concentration immediately after venous occlusion. Because arterial inflow remains unchanged while venous outflow is prevented, the increase in blood volume inside the tissue directly reflects the arterial blood entering the forearm. By calculating the slope of the HbT curve during the first few seconds of occlusion, the local blood flow can be estimated.

Arterial Occlusion of the Forearm

Arterial occlusion is achieved by inflating the cuff to a pressure above systolic blood pressure, thereby blocking both arterial inflow and venous outflow. Under these conditions, no fresh oxygenated blood can enter the tissue. However, cellular metabolism continues to consume oxygen, resulting in a decrease in HbO concentration and a corresponding increase in HbR

concentration over time. Because blood volume remains approximately constant during arterial occlusion, the observed changes in hemoglobin concentrations are primarily due to oxygen consumption by the tissue. These concentration changes can therefore be used to find local metabolic activity.

Measurement of Muscle Oxygen Consumption

Oxysoft provides continuous measurements of changes in HbO and HbR concentrations, allowing tissue oxygenation to be monitored in real time. During venous occlusion, the increase in blood volume provides information about blood flow, whereas during arterial occlusion, changes in HbO and HbR provide information about oxygen usage. Together, these measurements allow for a noninvasive measurement of both vascular and metabolic function in skeletal muscle.

In this experiment, a NIRS sensor was placed on the forearm of a volunteer to continuously monitor changes in HbO and HbR concentrations. A pressure cuff was used to perform both venous and arterial occlusions. During venous occlusion, arterial inflow was maintained while venous return was restricted, allowing forearm blood flow to be estimated from changes in total hemoglobin concentration. During arterial occlusion, both arterial inflow and venous outflow were blocked, allowing tissue oxygen consumption to be determined from changes in hemoglobin oxygenation. The resulting data was analyzed using Oxysoft data and Python to quantify physiological responses.

APPARATUS

A PortaMon wireless continuous-wave near-infrared spectroscopy (CW-NIRS) system was used to monitor changes in oxyhemoglobin (HbO) and deoxyhemoglobin (HbR) concentrations in the forearm muscle. The device uses light at 760 nm and 850 nm and transmits data wirelessly via Bluetooth to a computer for real-time acquisition. A blood pressure cuff was used to produce venous and arterial occlusions, while adjustable hand grips were used during exercise measurements. The NIRS sensor was secured to the forearm over the muscle of interest. The experimental setup is shown in Figure 4.

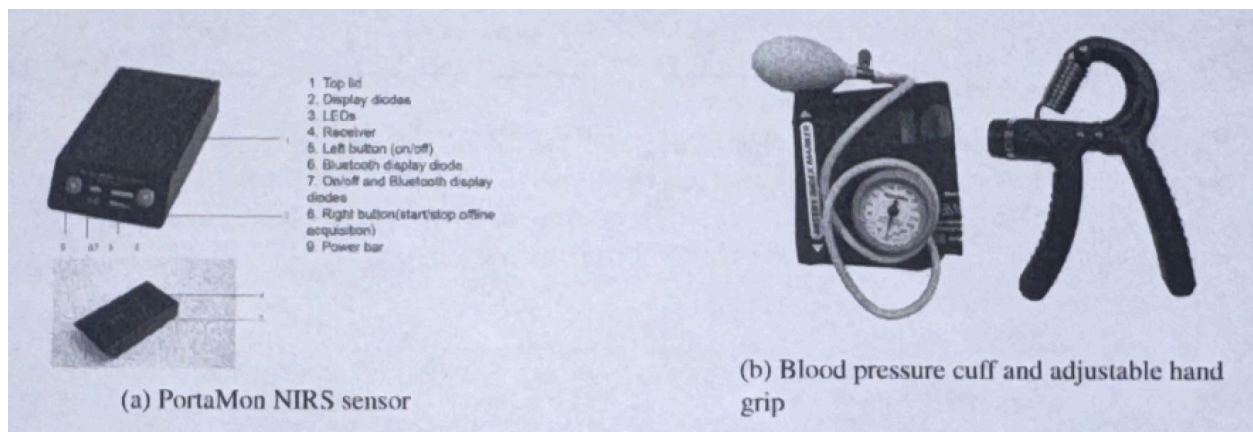


Figure 4¹: Experimental apparatus. (a) The PortaMon CW-NIRS device positioned on the upper forearm for measurement of HbO and HbR concentration changes. (b) The blood pressure cuff and adjustable hand grip used to induce venous and arterial occlusions during data collection.

PROCEDURE

Measurements were performed using the Oxysoft software (Artinis Medical Systems) on a laboratory laptop provided by the instructors. Data acquisition was carried out using a PortaMon

continuous-wave near-infrared spectroscopy (CW-NIRS) device.

Instrument Setup and Initialization

- The PortaMon device was powered on and allowed to stabilize for several minutes prior to data collection.
- Bluetooth connectivity between the PortaMon and the laptop was established.
- Oxysoft software was launched, and a new measurement was created using the **DAQ Measurement Wizard**.
- A new data file was generated, and the measurement was configured.
- The optode configuration (3-sensor template with fit factor) was selected.
- The differential path length factor (DPF) was automatically estimated using subject age via the built-in calculator.
- The sampling rate was set to **5 Hz**, providing sufficient temporal resolution for hemodynamic changes during occlusion.
- Select "Start measurement after finishing the wizard" and click "Finish."
- The PortaMon sensor was secured to the upper forearm using an elastic wrap and covered to minimize ambient light interference.
- A 2-minute baseline recording was collected prior to initiating any experimental trials.

Experimental Protocol

Two subjects were tested. Each trial consisted of a 60 s measurement period followed by a 60 s rest period (no cuff pressure or exercise) to allow partial physiological recovery.

The experimental conditions are summarized in Table 1.

Table 1: Experimental Trial Matrix

<i>Trial</i>	<i>50 mmHg (Venous)</i>	<i>200 mmHg (Arterial)</i>	<i>Gripping Intensity</i>
0	No	No	None (baseline)
1	✓	No	Low
2	No	✓	Low
3	✓	No	High
4	No	✓	High

Each subject completed all five trials in sequence.

Occlusion and Exercise Procedure

For each trial:

- The subject was seated comfortably with the PortaMon secured to the forearm.
- The blood pressure cuff was inflated to one of the following:
 - ◆ **50 mmHg** to induce **venous occlusion** (restricted venous return, arterial inflow maintained), or
 - ◆ **200 mmHg** to induce **arterial occlusion** (blocked arterial inflow and venous outflow).
- The subject performed isometric gripping using a standardized weighted device at either low or high intensity, depending on the trial condition.
- Data was recorded continuously for 60 s during each condition.
- After each trial, cuff pressure was released and gripping ceased, followed by 60s of rest.

Grip intensity was defined qualitatively due to the inability to directly measure force; however, a standardized gripping device was used to maintain consistency between trials.

SOFTWARE PROGRAMMING

While Oxysoft provides enough analysis capabilities for this lab, an automated pipeline was developed to streamline data parsing, plotting, and analysis. A custom Python analysis platform was developed consisting of two primary components: a graphical user interface (GUI), `PhysicsAnalysisGUI`¹⁰, and a reusable analysis library called `PhysicsLibrary`¹¹. The GUI handles user interaction, data visualization, marker placement, and export functionality, while the `PhysicsLibrary` module performs data loading, preprocessing, and numerical analysis. This separation improves code maintainability and allows analysis routines to be reused across multiple experiments.

Table 2: Summary of GUI Functions Used for Analysis

<i>Function / Module</i>	<i>File</i>	<i>Purpose</i>	<i>Input</i>	<i>Output</i>
load_dataset_file()	PhysicsLibrary ¹¹	Parses data files	Oxysoft .txt file path	HbO, HbR arrays, sample rate, events
simple_plot()	PhysicsAnalysisGUI ¹⁰	Renders HbO, HbR, HbT time series with per-channel and mean traces	Loaded cached object	Interactive matplotlib figure
toggle_marker_mode()	PhysicsAnalysisGUI ¹⁰	Enables click-to-place event markers on plot	User click at time t	Named vertical marker added to cache
launch_curve_fit()	PhysicsAnalysisGUI ¹⁰	Fits selected model to user-defined segment via two mouse clicks	Two time indices, source line	Fit parameters, R ² , overlay plot
launch_fft()	PhysicsAnalysisGUI ¹⁰	Computes FFT of HbO/HbR signals in a time window	Centre time, window length (s)	Power spectrum with annotated peaks
fit_model_to_segment()	PhysicsLibrary ¹¹	Wraps scipy.optimize.curve_fit for any registered model	x-y arrays, model function, p0 function	popt, R ² , fitted y values
linear_model(x, m, b)	PhysicsLibrary /models.py ¹¹	Linear model for slope extraction	x array, slope m, intercept b	$y = mx + b$

DATA AND CALCULATIONS

HbO and HbR concentration data was exported from Oxysoft as **.txt** files and imported into the custom Python analysis app (PhysicsAnalysisGUI¹⁰, PhysicsLibrary¹¹). Each file contains time-series measurements of HbO and HbR concentration changes (in μM) at 5 Hz. Event markers were placed manually on each trial to identify occlusion onset and offset.

Heart Rate Extraction via FFT

Resting heart rate was extracted from the HbO time series of the baseline (control) trial using a Fast Fourier Transform. A 15-second window centred on a stable baseline period was used for Subject 1, and a 10-second window for Subject 2. The dominant spectral peak in the HbO channel corresponds to the resting heart rate in bpm. The FFT frequency resolution is:

$$\Delta f = \frac{1}{T}$$

where T is the window duration.

For Subject 1 ($T = 15$ s):

$$\Delta f = 0.067 \text{ Hz} \approx \mathbf{4 \text{ bpm.}}$$

For Subject 2 ($T = 10$ s):

$$\Delta f = 0.10 \text{ Hz} \approx \mathbf{6 \text{ bpm.}}$$

Sample calculationSubject 1:

Dominant peak at $f = 0.87 \text{ Hz} \rightarrow$

$$HR = 0.87 \times 60 = \mathbf{52 \pm 4 \text{ bpm}}.$$

The first harmonic at 1.67 Hz ($\approx 100 \text{ bpm} = 2 \times 52$) confirms cardiac origin.

Subject 2:

Dominant cardiac peak at $f = 1.20 \text{ Hz} \rightarrow$

$$HR = 1.20 \times 60 = \mathbf{72 \pm 6 \text{ bpm}}.$$

The first harmonic at 2.40 Hz ($\approx 144 \text{ bpm} = 2 \times 72$) confirms cardiac origin.

A respiratory peak at 0.30 Hz (18 breaths/min) was also identified in the HbO data for Subject 2.

This is consistent with breathing-induced sensor motion during data collection.

Blood Flow Estimation from Venous Occlusion

During venous occlusion (50 mmHg), arterial inflow is maintained while venous return is blocked. Blood accumulates in the forearm, producing a linear increase in HbT. The rate of HbT increase is proportional to forearm blood flow. A linear fit was applied to the HbT signal during the stable occlusion window:

$$HbT(t) = m \cdot t + b$$

The slope m ($\mu\text{M/s}$) is used as the blood flow index.

Sample Calculation with Subject 2, Low Intensity

P1(153.2 s, 20.24 μM) and P2(188.4 s, 22.83 μM)

$$m = \frac{(22.83 - 20.24)}{(188.4 - 153.2)} = \frac{2.59}{35.2}$$

$$= 0.0708 \mu\text{M/s}$$

Oxygen Consumption Estimation from Arterial Occlusion

During arterial occlusion (200 mmHg), both arterial inflow and venous outflow are blocked.

Blood volume remains approximately constant ($\text{HbT} \approx \text{flat}$, $R^2 < 0.3$ in all trials), while oxygen is consumed by the tissue. This is reflected in a linear decrease in HbO and a corresponding increase in HbR:

$$VO_2 = - \frac{d[\text{HbO}]}{dt}$$

A linear fit was applied to the HbO signal during the stable occlusion window.

Sample calculation with Subject 2, Low Intensity:

P1(171.6 s, 5.32 μM) and P2(195.4 s, 4.71 μM)

$$m = \frac{4.71 - 5.32}{195.4 - 171.6} = \frac{-0.613}{23.8}$$

$$= -0.0248 \mu\text{M/s}$$

$$VO_2 = | -0.0248 | = 0.0248 \mu\text{M/s}$$

The HbT slope for the same trial was $0.0355 \mu\text{M/s}$, $R^2 = 0.982$ which confirms that the blood volume was approximately constant, validating the arterial occlusion assumption.

Uncertainty: Slope uncertainties were estimated from the diagonal elements of the covariance matrix returned by the linear regression, yielding values of ± 0.003 – $0.010 \mu\text{M/s}$ for venous occlusion trials and ± 0.0003 – $0.003 \mu\text{M/s}$ for arterial occlusion trials. These small values indicate that the fits are stable and that the dominant source of uncertainty is physiological noise, particularly the non-sinusoidal cardiac rhythm, which introduces signal variability that reduces R^2 below 0.90 in some trials. Additional unmeasurable sources of uncertainty include sensor motion during isometric gripping and individual anatomical factors such as subcutaneous fat thickness and skin pigmentation (or tattoos), which affect light penetration depth and optical path length.

RESULTS

Baseline and Heart Rate

Table 3: Resting Heart Rate For Both Subjects

<i>Subject</i>	<i>Dominant Peak (Hz)</i>	<i>Heart Rate (bpm)</i>	<i>Harmonic (Hz)</i>	<i>FF (%)</i>
Subject 1	0.87	52 ± 4	1.67 (100 bpm)	99.4
Subject 2	1.20	72 ± 6	2.40 (144 bpm)	99.9

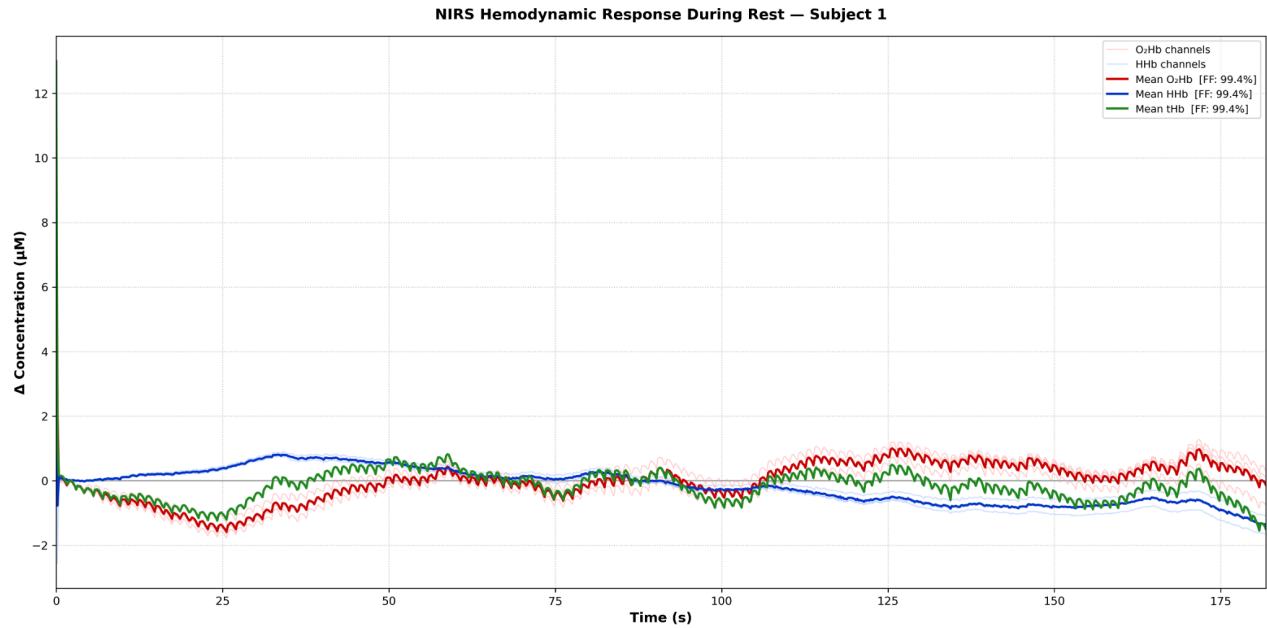


Figure 5: NIRS hemodynamic response during rest for Subject 1. Changes in oxyhemoglobin (HbO, red), deoxyhemoglobin (HbR, blue), and total hemoglobin (HbT, green) concentration are shown over the full recording duration. The fit factor (FF = 99.4%) indicates good optical contact with the skin.

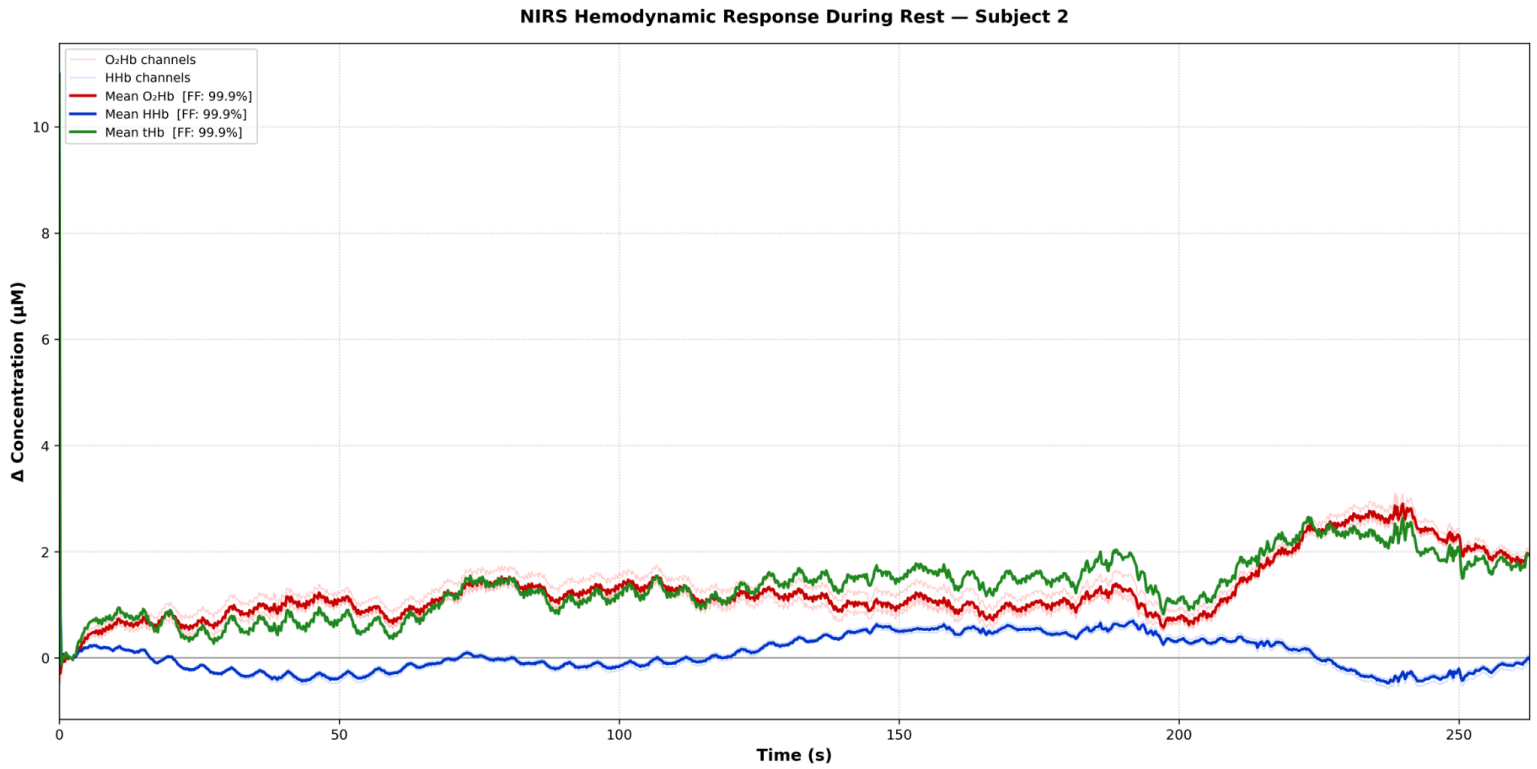


Figure 6: NIRS hemodynamic response during rest for Subject 2. HbO, HbR, and HbT concentration changes are shown over the full recording duration (FF = 99.9%).

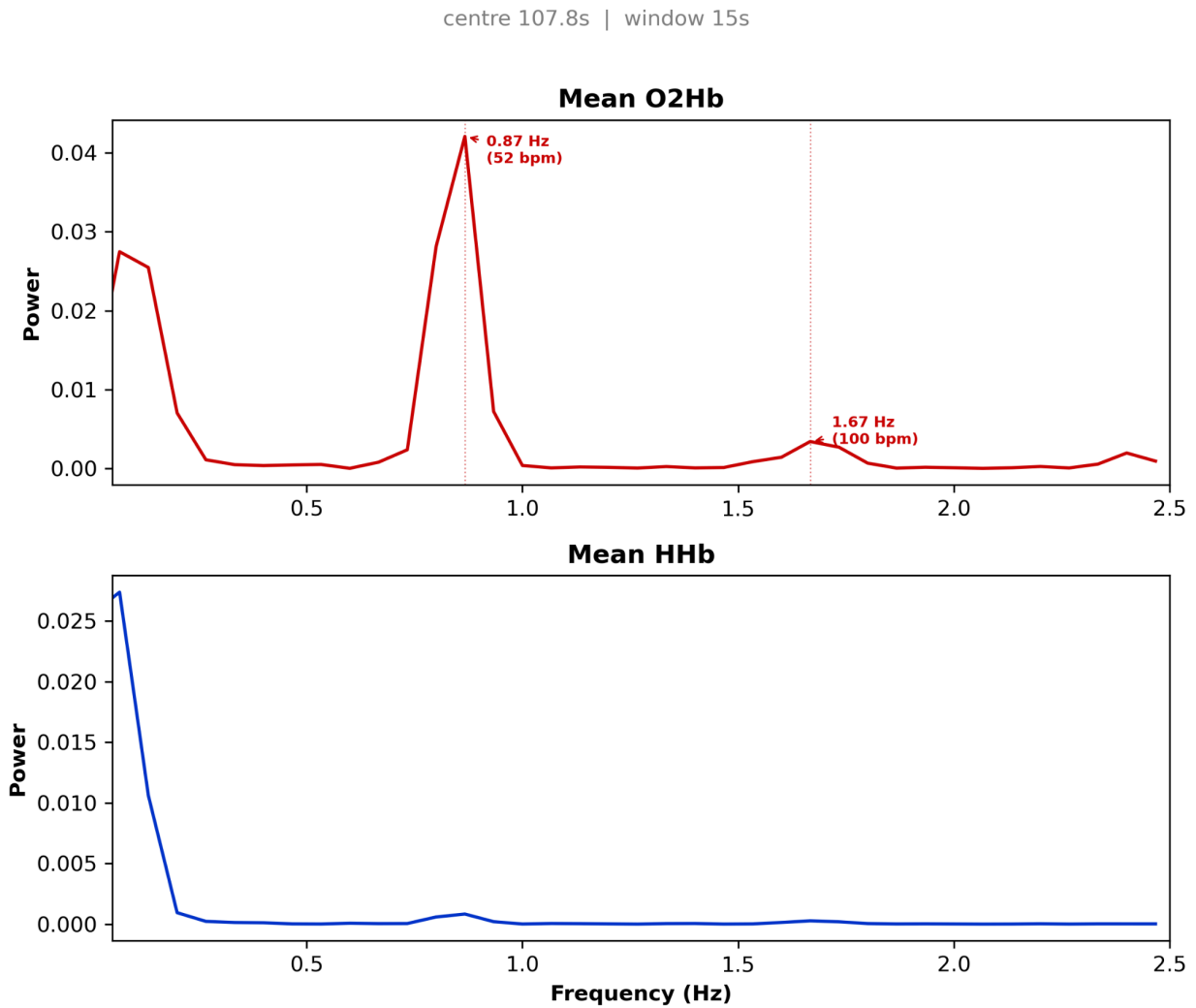


Figure 7: Frequency spectrum of the resting NIRS signal for Subject 1, computed from a 15-second window of the HbO baseline. The dominant peak at 0.87 Hz (52 bpm) corresponds to the cardiac pulse rate. The first harmonic at 1.67 Hz (100 bpm) confirms the cardiac origin of the signal.

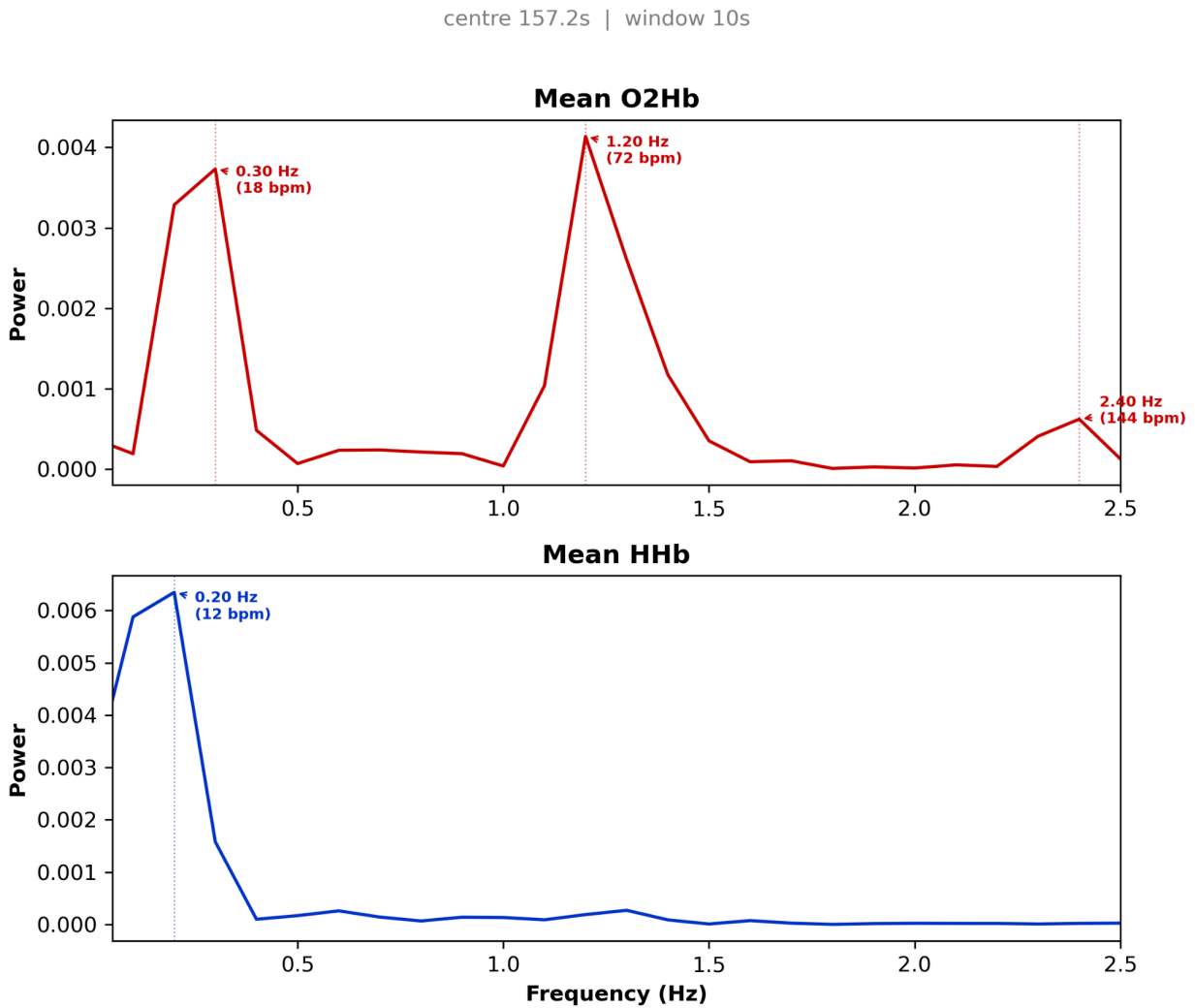


Figure 8: Frequency spectrum of the resting NIRS signal for Subject 2, computed from a 10-second window of the HbO baseline. The dominant cardiac peak appears at 1.20 Hz (72 bpm). A respiratory peak at 0.30 Hz (18 breaths/min) is visible in the HbO channel, attributed to breathing-induced oscillations and sensor motion.

Figures 5 and 6 show the resting NIRS signals for Subject 1 and Subject 2, respectively. Both subjects displayed stable HbO and HbR signals near zero during the baseline period, with small artifacts due to the 5 Hz sampling rate and possibly breathing-related motion artifacting. The mean fit factor was 99.4% for Subject 1 and 99.9% for Subject 2, indicating excellent agreement with the predicted modified Beer-Lambert model.

Figures 7 and 8 show the FFT power spectra of the resting HbO signal for each subject. Subject 1 showed a dominant cardiac peak at 0.87 Hz (52 bpm) with a first harmonic at 1.67 Hz (100 bpm), confirming the cardiac origin of the signal. Subject 2 showed a dominant cardiac peak at 1.20 Hz (72 bpm) with a first harmonic at 2.40 Hz (144 bpm), confirming the cardiac origin of the signal, along with a peak at 0.30 Hz (18 breaths/min) in the HbO channel, likely due to breathing-induced motion of the sensor during recording.

Venous Occlusion

Blood Flow

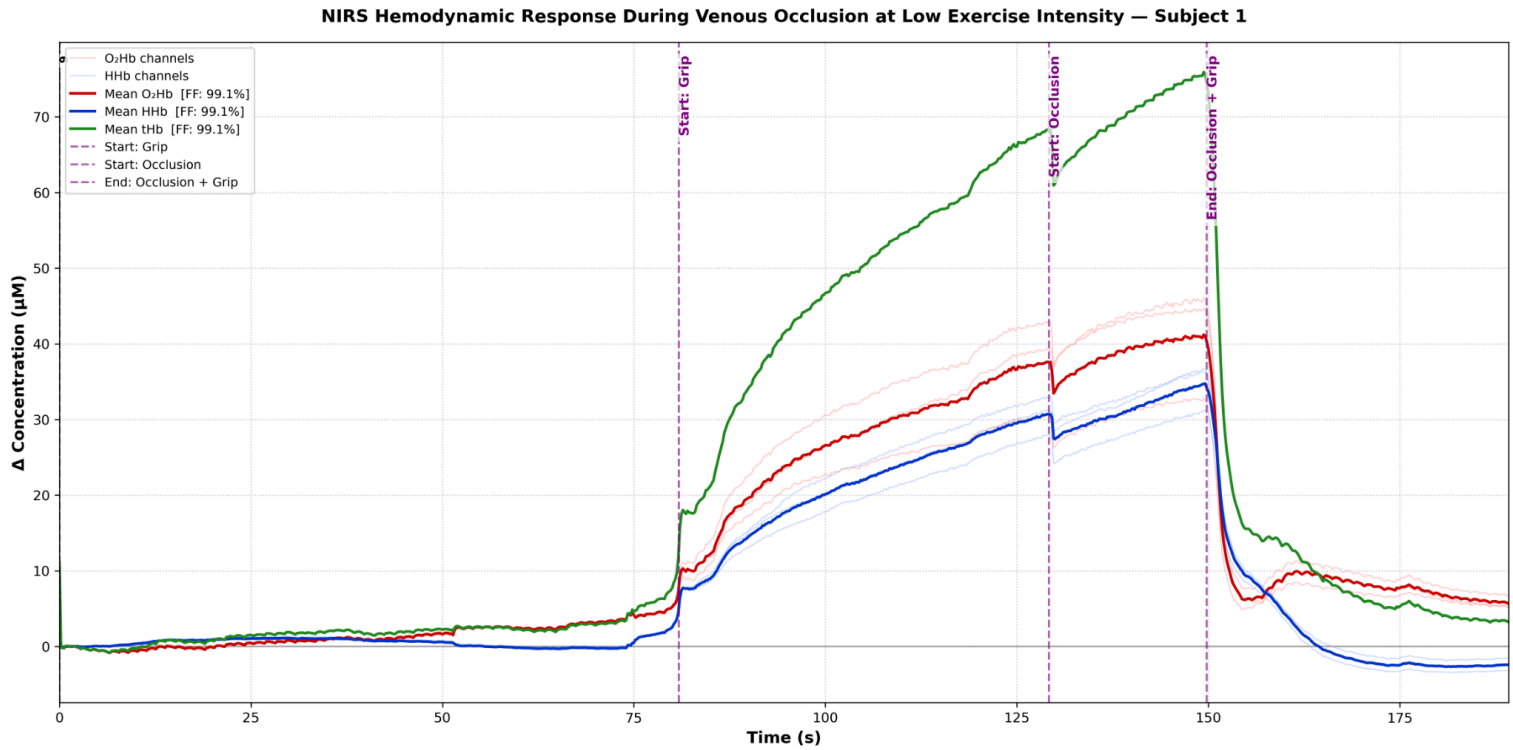


Figure 9: NIRS hemodynamic response during venous occlusion at low exercise intensity for Subject 1 (cuff pressure: 50 mmHg, FF = 99.1%). Event markers indicate grip start, occlusion start, and both their endings. A linear increase in HbT during occlusion reflects arterial blood accumulation.

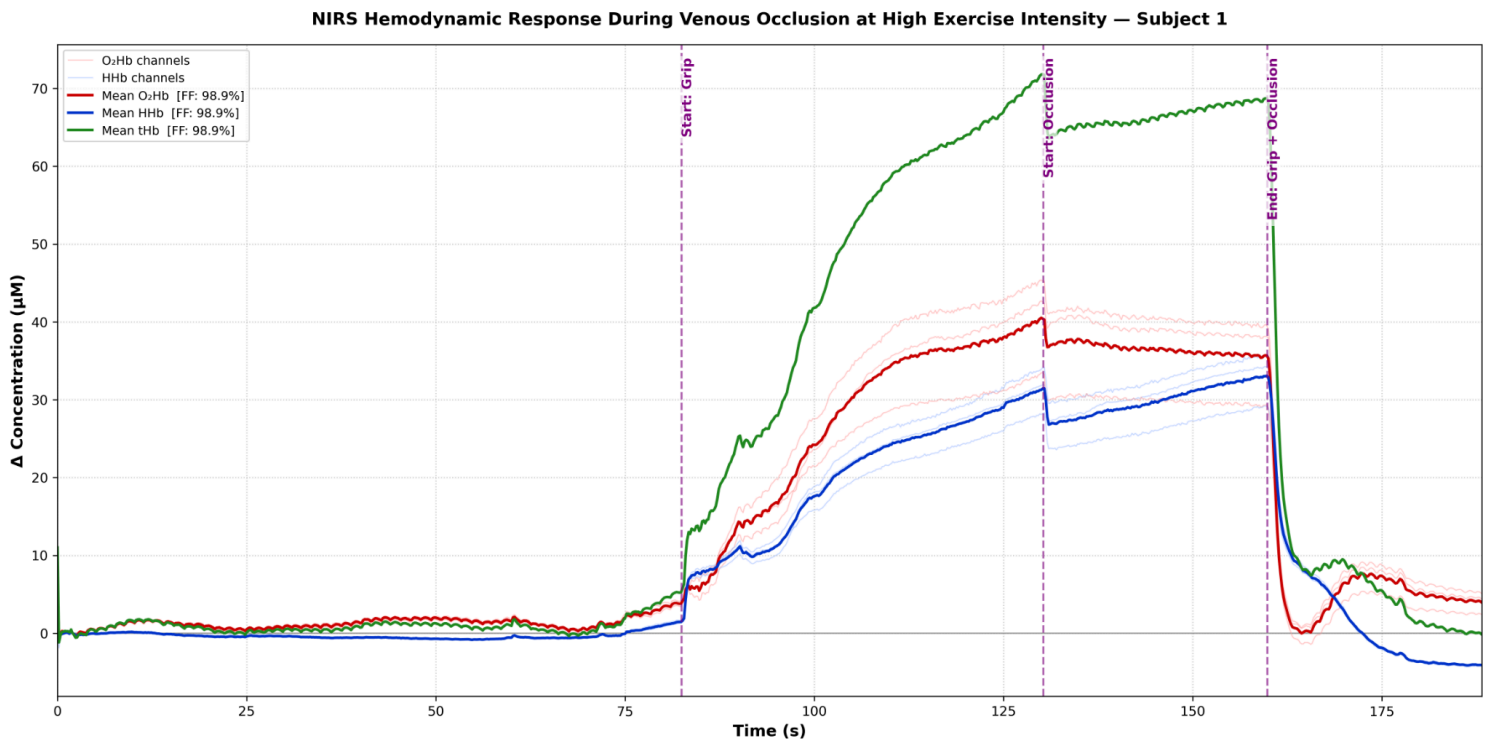


Figure 10: NIRS hemodynamic response during venous occlusion at high exercise intensity for Subject 1 (cuff pressure: 50 mmHg, FF = 98.9%). Higher exercise intensity produces a shallower HbT slope, inconsistent with the blood flow relationship to exercise intensity.

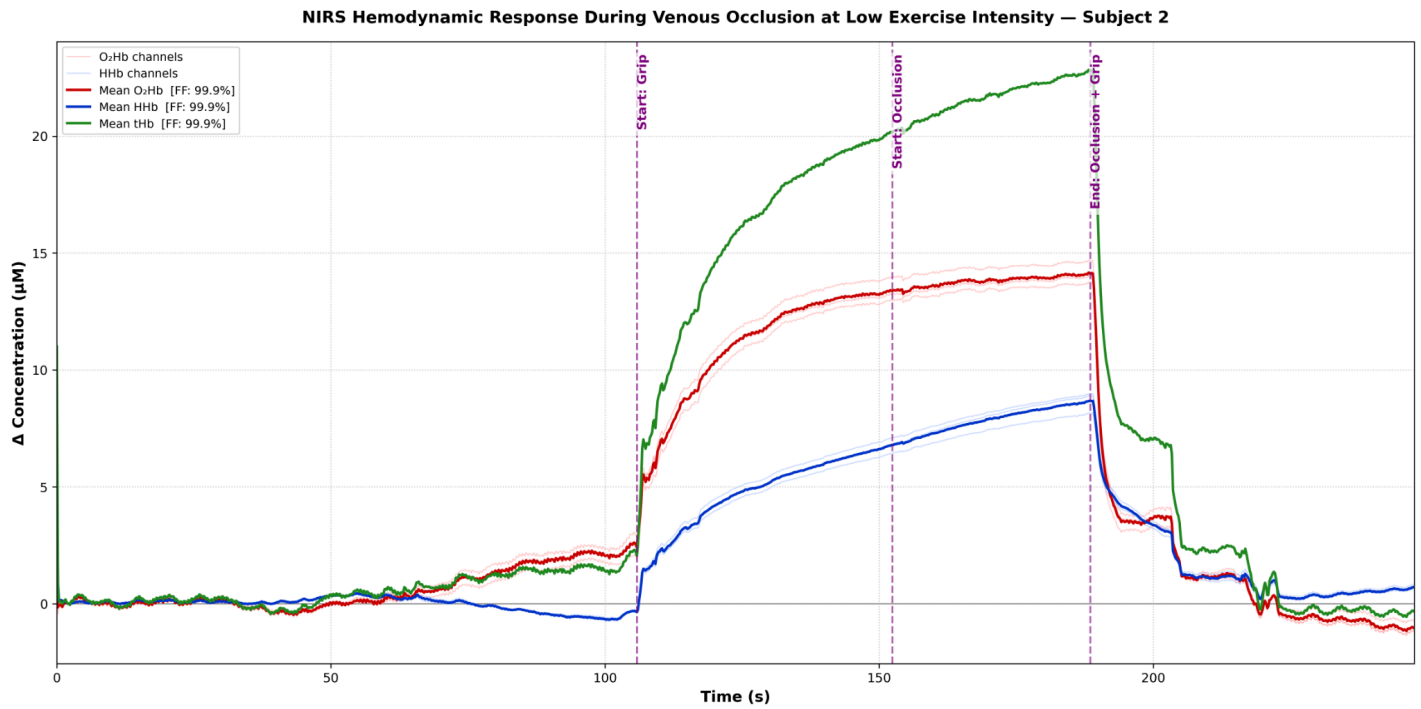


Figure 11: NIRS hemodynamic response during venous occlusion at low exercise intensity for Subject 2 (cuff pressure: 50 mmHg, FF = 99.9%).

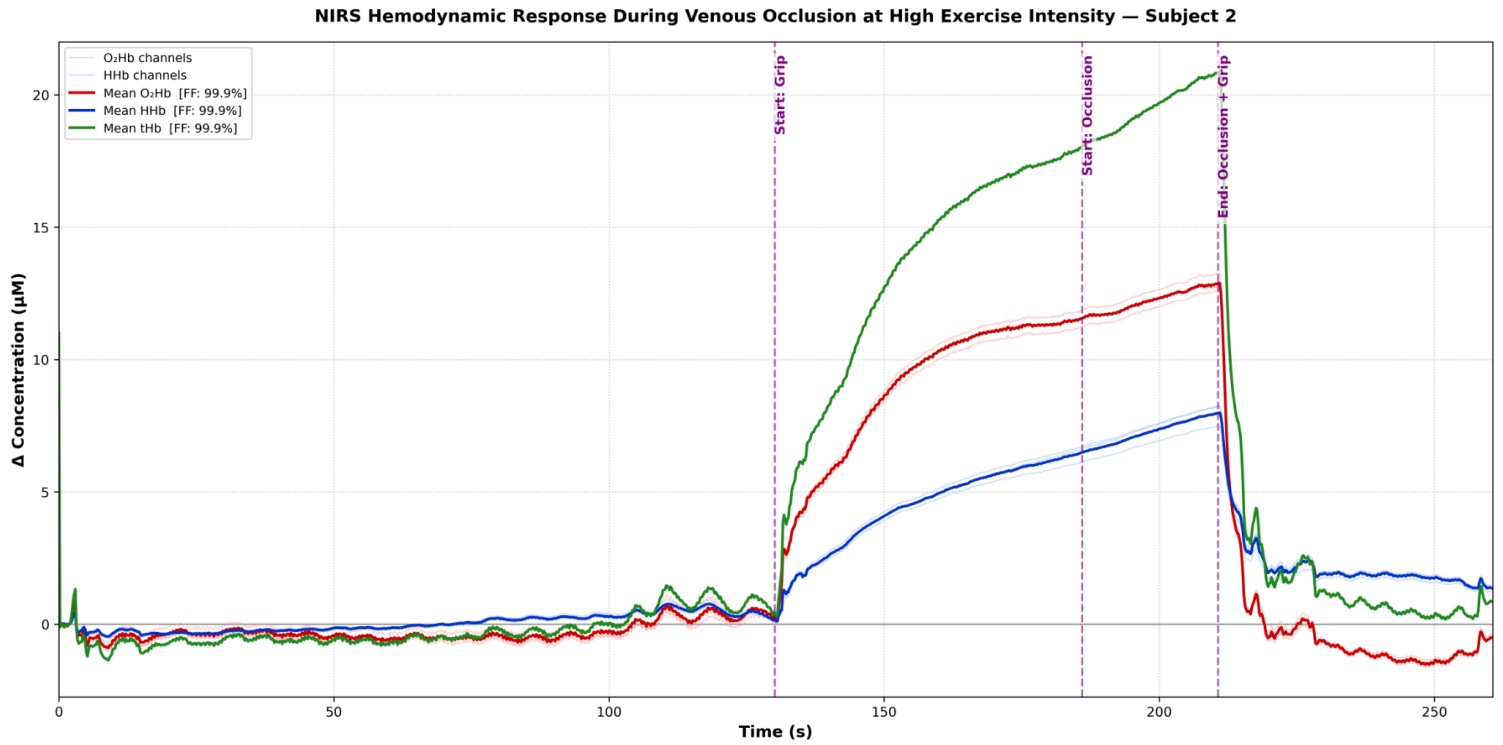


Figure 12: NIRS hemodynamic response during venous occlusion at high exercise intensity for Subject 2 (cuff pressure: 50 mmHg, FF = 99.9%).

Curve Fits

Table 4: HbT Linear Slope During Venous Occlusion (Blood Flow Index)

<i>Subject</i>	<i>Condition</i>	<i>HbT Slope</i> ($\mu\text{M/s}$)	R^2	Δt (s)
Subject 1	Low Intensity	0.690 ± 0.010	0.979	19.0
Subject 1	High Intensity	0.154 ± 0.003	0.959	28.2
Subject 2	Low Intensity	0.0708 ± 0.0009	0.975	35.2
Subject 2	High Intensity	0.1207 ± 0.0008	0.994	24.4

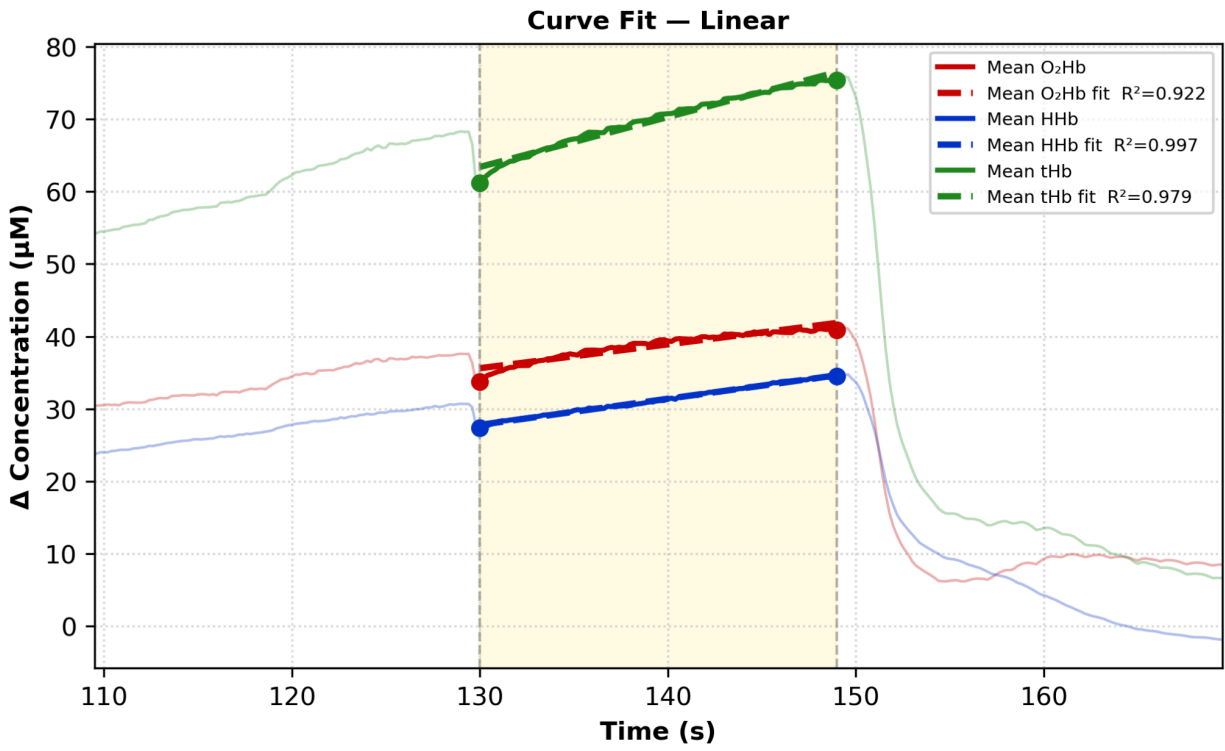


Figure 13: Linear curve fit of HbO, HbR, and HbT during the venous occlusion window at low exercise intensity for Subject 1. The shaded region indicates the fitted segment. The HbT slope

$m = 0.690 \pm 0.010 \text{ } \mu\text{M/s}$, $R^2 = 0.979$ are used as the blood flow index.

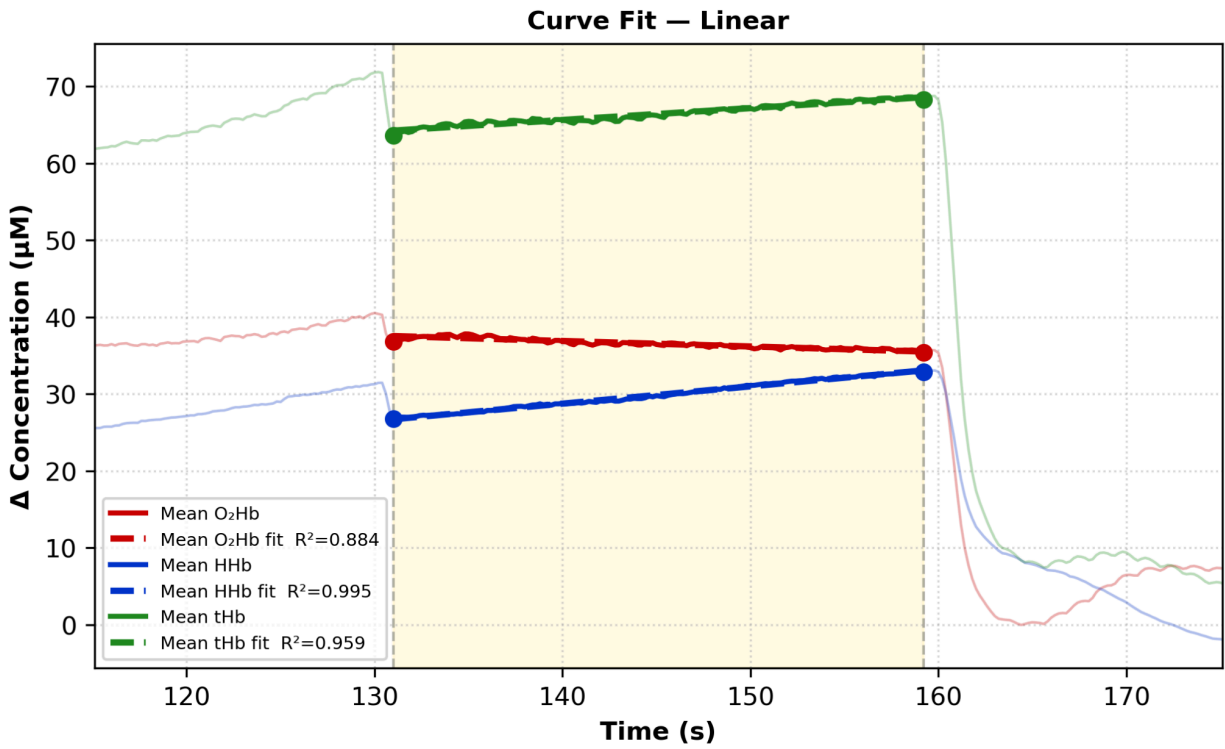


Figure 14: Linear curve fit during venous occlusion at high exercise intensity for Subject 1. The

HbT slope ($m = 0.154 \pm 0.003 \mu\text{M/s}$, $R^2 = 0.959$) is lower at low intensity, inconsistent with increased activity level.

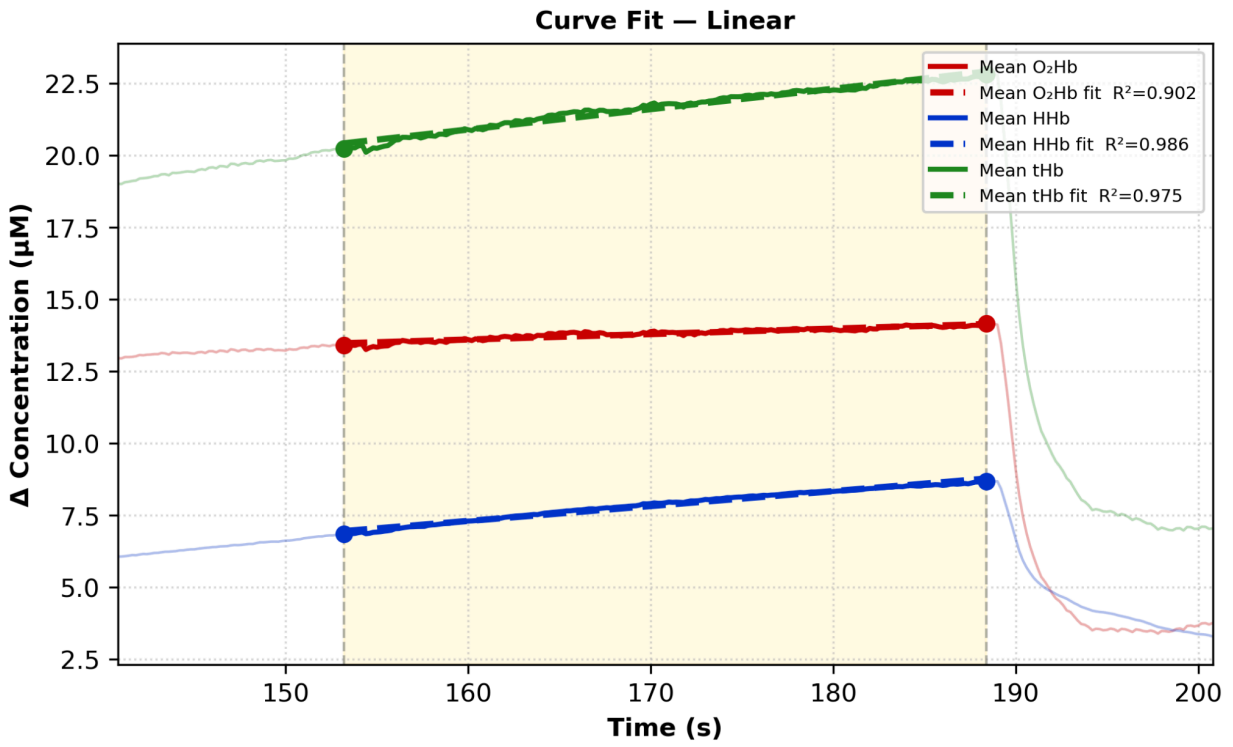


Figure 15: Linear curve fit during venous occlusion at low exercise intensity for Subject 2 (HbT

slope: $m = 0.0708 \pm 0.0009 \mu\text{M/s}$, $R^2 = 0.975$)

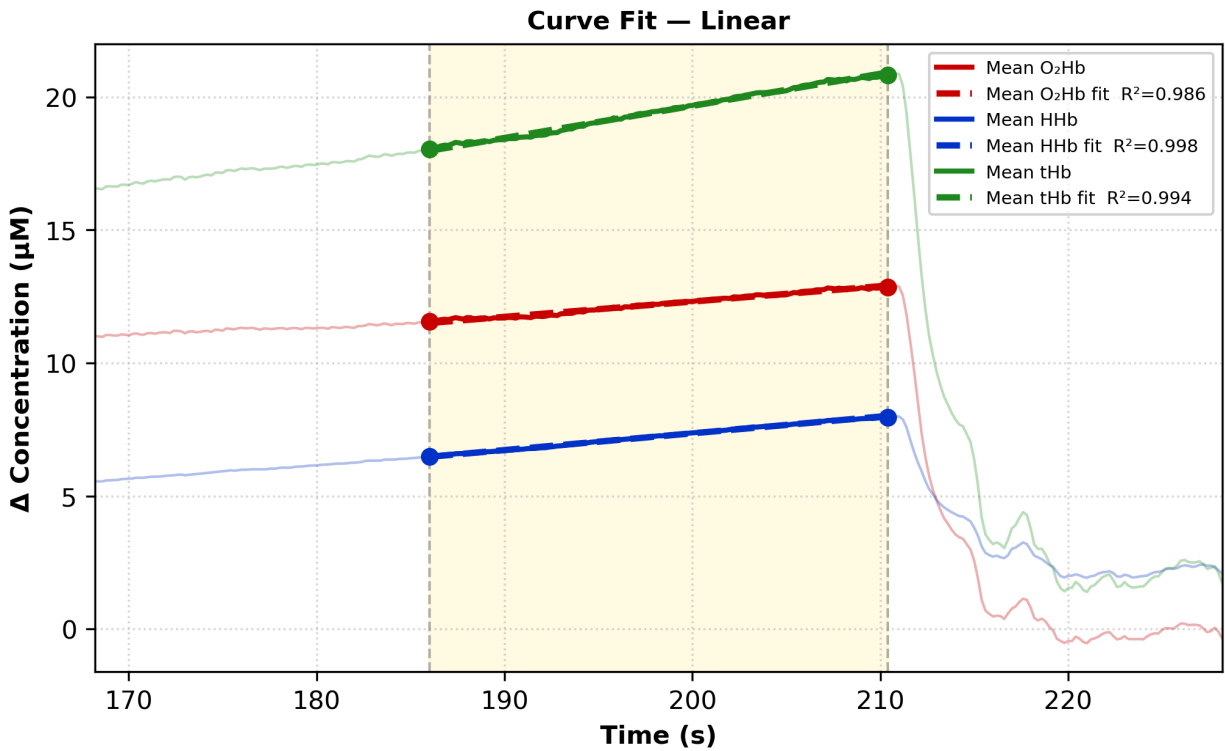


Figure 16: Linear curve fit during venous occlusion at high exercise intensity for Subject 2 (HbT slope: $m = 0.1207 \pm 0.0008 \mu\text{M/s}$, $R^2 = 0.994$)

Figures 9–12 show the full NIRS hemodynamic traces for venous occlusion trials. In all trials, HbT increased linearly following cuff inflation, confirming effective venous occlusion. HbO and HbR increased steadily. After cuff release, reactive hyperemia was observed, which is a characteristic rapid overshoot in HbO and HbT as blood flow resumes.

Figures 13–16 show the corresponding linear curve fits during the occlusion window. Fit quality was high for HbT and HbR ($R^2 > 0.92$ in most trials), while HbO fits showed slightly lower R^2 due to competing effects of blood accumulation and oxygen extraction. Although Subject 1 showed a negative relation between blood flow and isometric hold intensity, Subject 2's results follow theory.

Arterial Occlusion

Oxygen Consumption

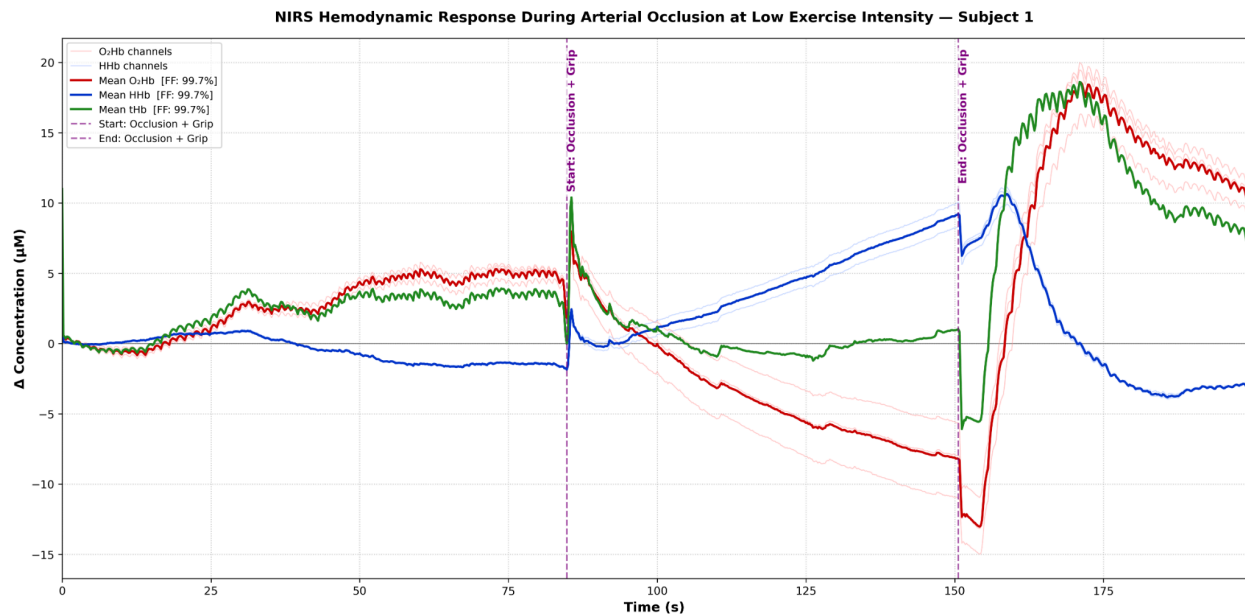


Figure 17: NIRS hemodynamic response during arterial occlusion at low exercise intensity for Subject 1 (cuff pressure: 200 mmHg, FF = 99.7%). HbO decreases and HbR increases monotonically during occlusion, reflecting tissue oxygen consumption. HbT remains approximately constant, confirming blood volume conservation.

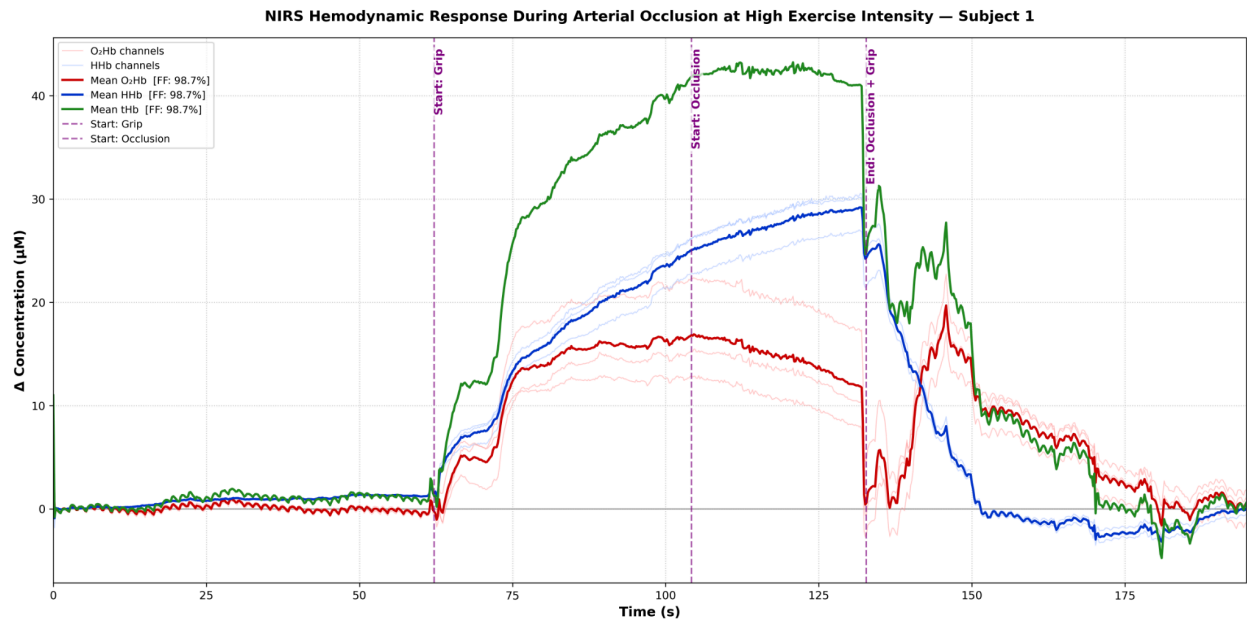


Figure 18: NIRS hemodynamic response during arterial occlusion at high exercise intensity for Subject 1 (cuff pressure: 200 mmHg, FF = 98.7%).

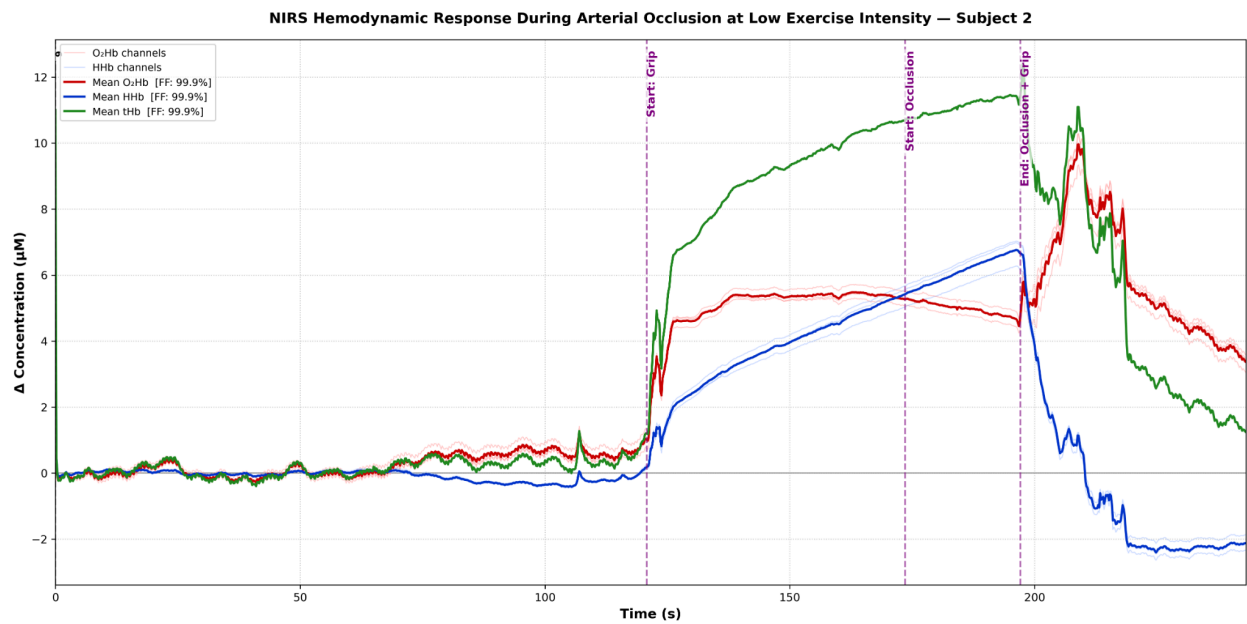


Figure 19: NIRS hemodynamic response during arterial occlusion at low exercise intensity for Subject 2 (cuff pressure: 200 mmHg, FF = 99.9%).

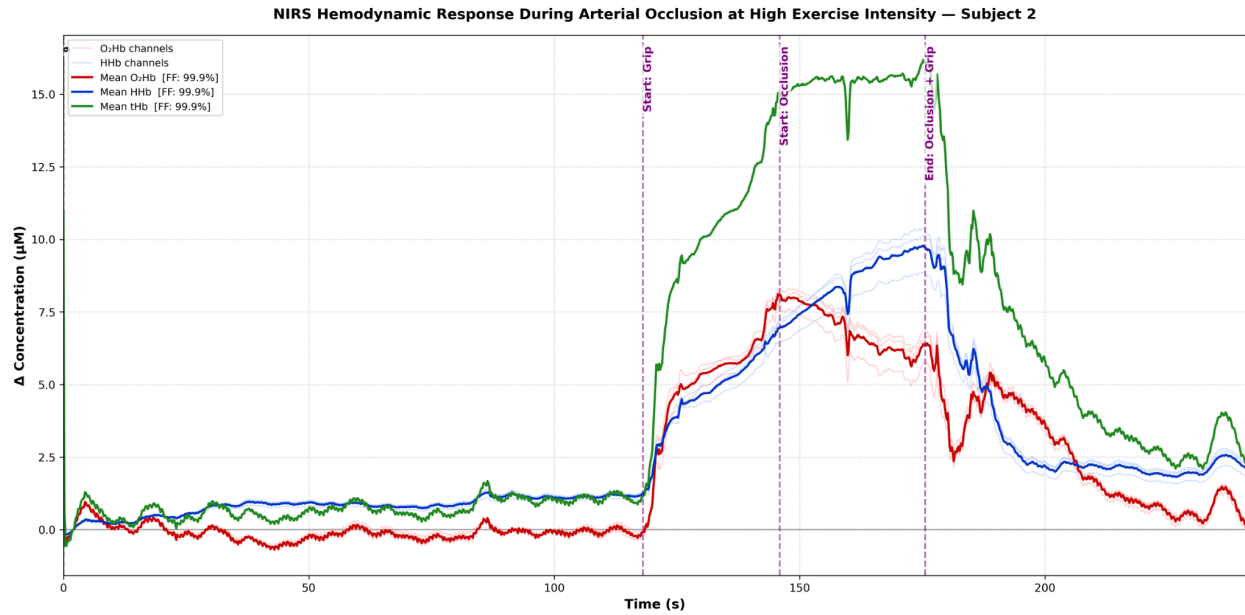


Figure 20: NIRS hemodynamic response during arterial occlusion at high exercise intensity for Subject 2 (cuff pressure: 200 mmHg, FF = 99.9%).

Curve Fits

Table 5: HbO Linear Slope During Arterial Occlusion (VO₂ Index)

<i>Subject</i>	<i>Condition</i>	<i>HbO Slope (μM/s)</i>	<i>HbR Slope (μM/s)</i>	<i>R² (HbO)</i>	<i>Δt (s)</i>
Subject 1	Low Intensity	-0.1458 ± 0.0016	$+0.1642 \pm 0.0005$	0.971	47.6
Subject 1	High Intensity	-0.1845 ± 0.0031	$+0.1500 \pm 0.0023$	0.965	25.6
Subject 2	Low Intensity	-0.0248 ± 0.0003	$+0.0603 \pm 0.0002$	0.982	23.8
Subject 2	High Intensity	-0.0781 ± 0.0024	$+0.0932 \pm 0.0023$	0.875	29.8

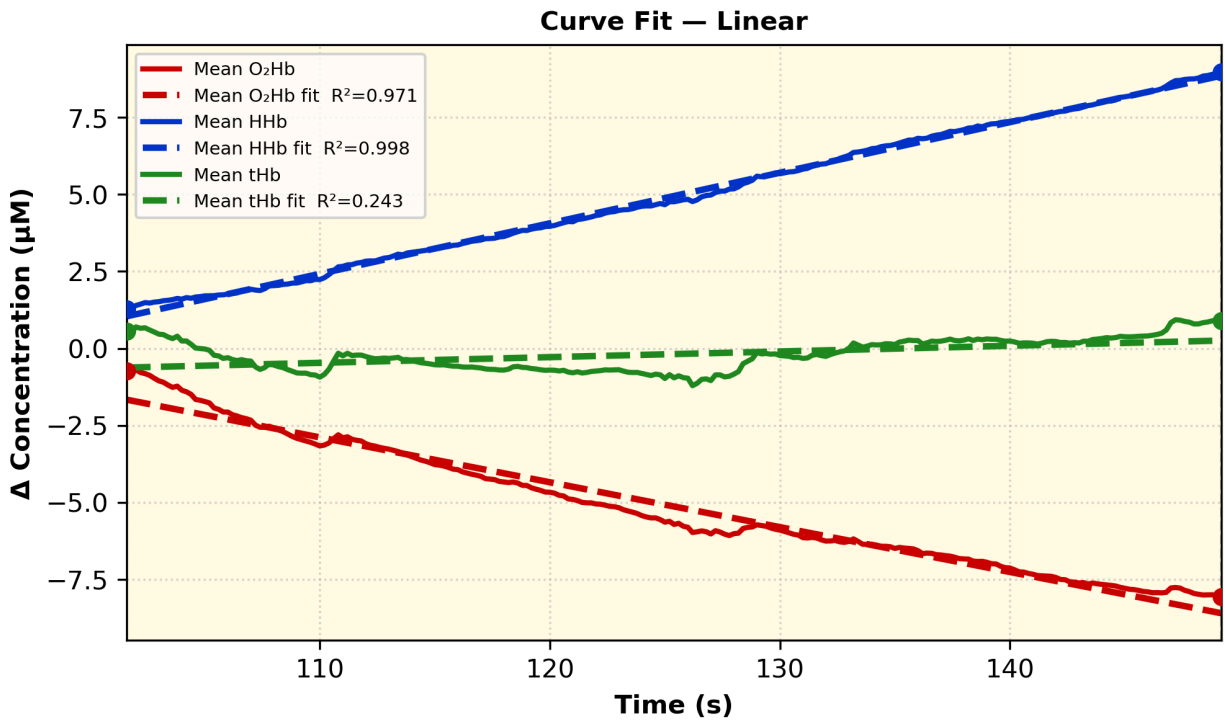


Figure 21: Linear curve fit of HbO and HbR during arterial occlusion at low exercise intensity for Subject 1. The HbO slope ($m = -0.1458 \pm 0.0016 \text{ } \mu\text{M/s}$, $R^2 = 0.971$) provides the VO_2 index. The HbT fit ($R^2 = 0.243$) confirms blood volume was approximately constant during occlusion.

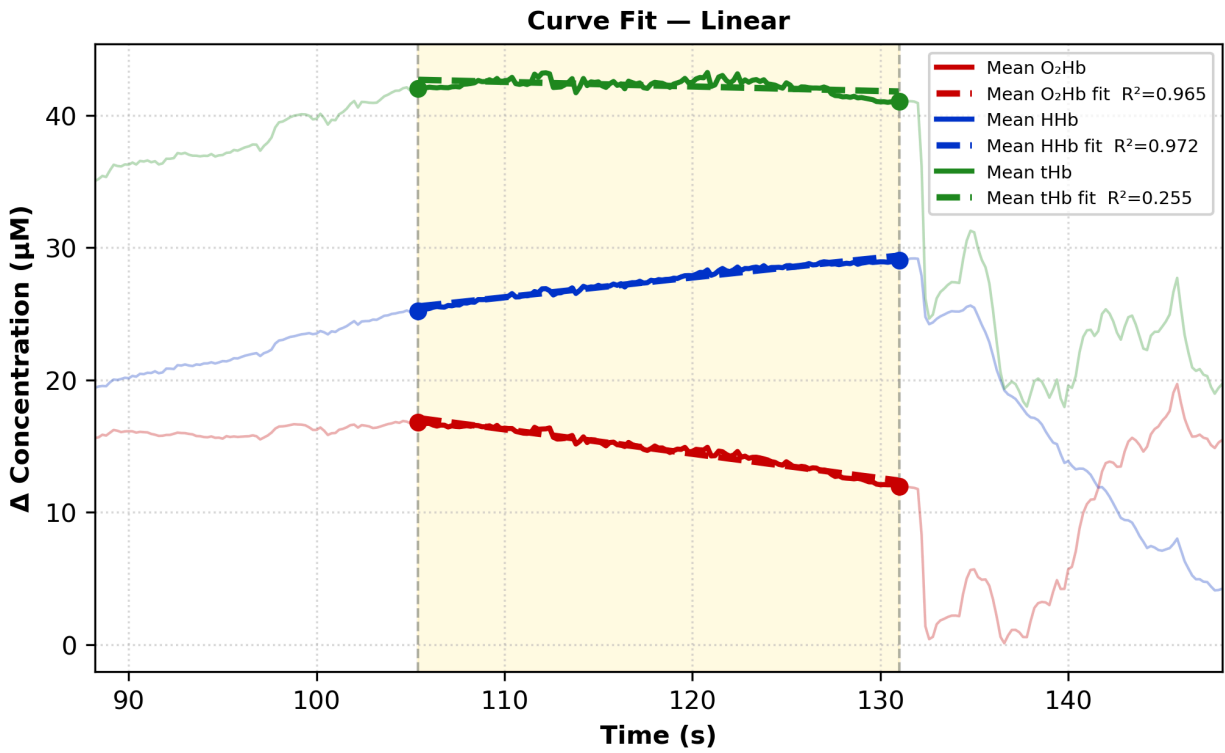


Figure 22: Linear curve fit during arterial occlusion at high exercise intensity for Subject 1 (HbO slope: $m = -0.1845 \pm 0.0031 \mu\text{M/s}$, $R^2 = 0.965$). The slope represents the VO_2 index, and the HbT fit ($R^2 = 0.255$) confirms total blood volume was constant during occlusion.

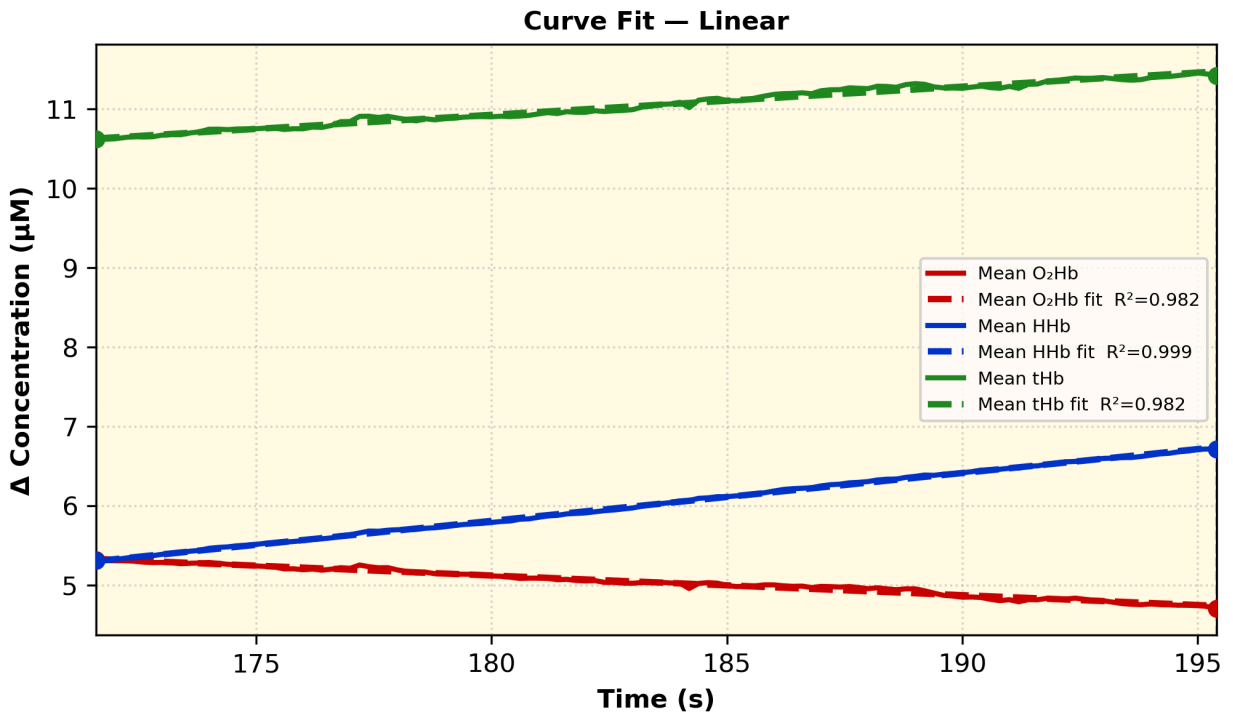


Figure 23: Linear curve fit during arterial occlusion (200 mmHg) at low exercise intensity for Subject 2 (HbO slope: $m = -0.0248 \pm 0.0003 \mu\text{M/s}$, $R^2 = 0.982$). Occlusion is not reached, as seen with the positive HbT fit.

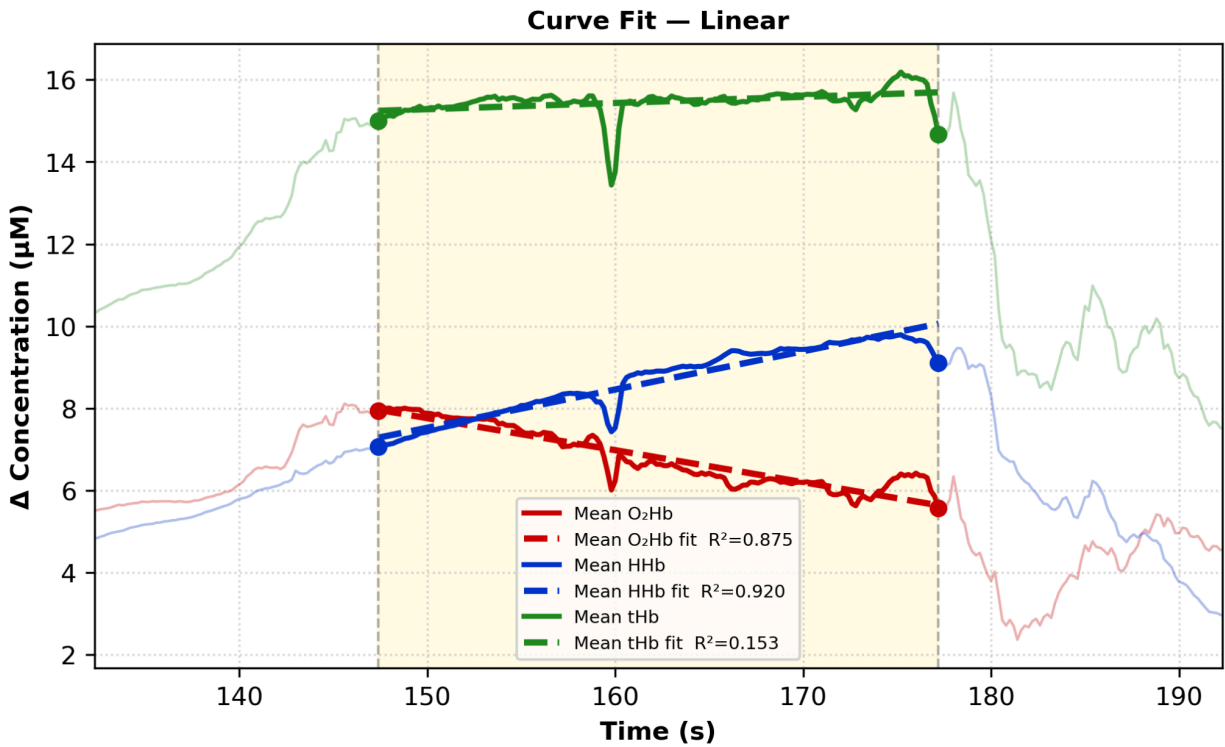


Figure 24: Linear curve fit during arterial occlusion at high exercise intensity for Subject 2 (HbO slope: $m = -0.0781 \pm 0.0024 \mu\text{M/s}$, $R^2 = 0.875$). Occlusion is reached as seen with the poor HbT fit. A few artifacts are visible.

Figures 17–20 show the arterial occlusion trials. In all cases, HbO decreased and HbR increased monotonically during the occlusion period, consistent with continued oxygen consumption in the absence of arterial supply. The HbT signal remained approximately flat ($R^2 < 0.30$ in all arterial trials), confirming that blood volume was conserved and that the observed hemoglobin changes reflect metabolic activity rather than blood volume shifts. Figures 21–24 show the HbO linear fits during the occlusion window. $\text{VO}_2 \left(\left| \frac{d[\text{HbO}]}{dt} \right| \right)$ increased with exercise intensity for both subjects. Subject 1 showed substantially higher oxygen consumption rates than Subject 2 at both intensities, suggesting possibly higher forearm muscle mass.

DISCUSSION

Heart Rate

Resting heart rates of 52 bpm (Subject 1) and 72 bpm (Subject 2) were obtained from FFT analysis of the NIRS baseline signal. Both values fall within the normal adult resting heart rate range of 60–100 bpm, though Subject 1's value is slightly below this range, which is not uncommon in physically active individuals. The cardiac signal was clearly identifiable as the dominant spectral peak in the HbO channel, with a first harmonic at exactly twice the fundamental frequency confirming its cardiac origin. The HbR channel showed weaker cardiac modulation for both subjects, consistent with the known lower sensitivity of HbR to arterial pulsatility compared to HbO. Subject 2's FFT also showed a respiratory peak at 0.30 Hz (18 breaths/min), which is a well-documented phenomenon in NIRS¹²; breathing modulates intrathoracic pressure and venous return, producing real hemodynamic oscillations visible in the NIRS signal in addition to any motion artifact. The respiratory peak is as large as the cardiac peak in the HbR channel, consistent with HbR's great sensitivity to venous volume changes driven by breathing.

Blood Flow

Blood flow indices derived from the HbT slope during venous occlusion showed different trends between subjects. Subject 2 behaved as expected, with the HbT slope increasing from 0.071 $\mu\text{M/s}$ at low intensity to 0.121 $\mu\text{M/s}$ at high intensity, consistent with greater arterial blood supply to meet higher metabolic demand. Subject 1, however, showed the opposite trend: the HbT slope decreased from 0.690 $\mu\text{M/s}$ at low intensity to 0.154 $\mu\text{M/s}$ at high intensity. This is inconsistent with the expected exercise-intensity relationship and is likely attributable to

high-force isometric contraction compressing the intramuscular vasculature, partially restricting arterial inflow even during venous occlusion. At sufficiently high grip forces, mechanical compression of vessels can counteract the vessel dilation response to exercise, reducing net blood accumulation during the occlusion window and thus underestimating true blood flow¹³.

Alternatively, the low-intensity trial for Subject 1 may have coincided with a period of elevated baseline vasodilation, inflating that slope relative to the high-intensity trial. Despite this anomaly, Subject 1's absolute HbT slopes were substantially higher than Subject 2's across both conditions, suggesting greater forearm oxygen demand overall, possibly reflecting higher absolute grip force output given that intensity was defined qualitatively rather than quantitatively.

The linear fit quality for HbT was excellent ($R^2 \geq 0.959$) in most trials, supporting the assumption of constant arterial inflow during the occlusion window. The slight deviation from linearity observed in some trials, particularly at high intensity, reflects the competing effect of continued oxygen extraction reducing HbO while HbR rises, which introduces mild curvature into the HbT signal.

Oxygen Consumption

VO₂ estimates from the HbO slope during arterial occlusion increased with exercise intensity for both subjects, consistent with the expected increase in metabolic demand during higher-intensity isometric contraction. Subject 1 showed approximately 2–6 times higher VO₂ than Subject 2 across conditions, mirroring the blood flow differences and suggesting Subject 1 either performed at greater absolute intensity or had higher muscle oxidative capacity.

The HbT signal remained flat ($R^2 < 0.30$) during all arterial occlusion trials, confirming that the

observed HbO and HbR changes reflect tissue oxygen consumption rather than blood volume shifts. This validates the core assumption of the arterial occlusion method.

The HbR slope was consistently larger in magnitude than the HbO slope for Subject 2, while they were more comparable for Subject 1. This asymmetry in Subject 2 may reflect a small residual blood volume change during occlusion. A possible systematic error is imperfect arterial occlusion; if cuff pressure at 200 mmHg did not fully occlude arterial inflow (possible in subjects with higher systolic pressure), some fresh blood could continue entering the tissue, causing HbO to decrease more slowly than true VO_2 would predict, thus leading to underestimates in oxygen consumption.

Other sources of error include motion artifact from isometric gripping, which introduces noise into the NIRS signal and reduces R^2 values (notably $R^2 = 0.875$ for Subject 2 high intensity). The finite 5 Hz sampling rate also limits the ability to resolve rapid hemodynamic transitions at cuff inflation and release. Formal uncertainties in the fitted slopes, derived from the covariance matrix of the linear regression, were small (± 0.001 – 0.010 $\mu\text{M/s}$ for venous and ± 0.0003 – 0.003 $\mu\text{M/s}$ for arterial trials), indicating that the dominant source of variability is physiological noise rather than fitting.

Comparison to Expected Values

Literature values for resting forearm VO_2 measured by NIRS-occlusion in healthy adults typically range from 0.010 to 0.050 $\mu\text{M/s}$ and increase drastically during moderate exercise^{7,8}. Subject 2's values (0.025–0.078 $\mu\text{M/s}$) are consistent with this range. Subject 1's values (0.146–0.185 $\mu\text{M/s}$) are higher than typical resting values but possible during high-intensity isometric contraction. Forearm blood flow at rest is typically reported around 2–5 mL/min/100g

tissue⁷, and the HbT slopes observed here are consistent with a moderate to high exercise state for both subjects.

Overall, the experiment successfully demonstrated the expected physiological responses to venous and arterial occlusion, with clear increases in both blood flow and oxygen consumption with exercise intensity, validating the NIRS-occlusion methodology for non-invasive assessment of skeletal muscle hemodynamics.

REFERENCES

1. Grova, Christophe, and Patrick Doane. Laboratory Manual - Experimental Physics II.

Concordia University, Jan. 2023, pp. 39–48.
2. Ye, X., Doi, T., Arakawa, O., et al. A novel spatially resolved interactance spectroscopy system to estimate the degree of red coloration in red-fleshed apples. *Sci Rep* 11, 21982 (2021). <https://doi.org/10.1038/s41598-021-01468-z>
3. Martin Buchheit, Pierre Ufland, Bachar Haydar, Paul B. Laursen, and Said Ahmaidi.

Reproducibility and sensitivity of muscle reoxygenation and oxygen uptake recovery kinetics following running exercise in the field. *Clinical Physiology and Functional Imaging*, 31(5):337-346, 2011.
4. Catherine Mary Hesford, Stewart J. Laing, Marco Cardinale, and Chris E. Cooper.

Asymetry of quadriceps muscle oxygenation during elite short-track speed skating. *Medicine & Science in Sports & Medicine*, 193(6):501-508, 2012.
5. Kevin G. Thompson, Louise Turner, Jonathon Prichard, Fiona Dodd, David O. Kennedy, Crystal Haskell, James R. Blackwell, and Andrew M. Jones. Influence of dietary nitrate supplementation on physiological and cognitive responses to incremental cycle exercise. *Respiratory Physiology & Neurobiology*, 193:11-20, 2014.
6. Jong-Shyan Wang, Tieh-Cheng Fu, Hen-Yu Lien, Chao-Hung Wang, Chih-Chin Hsu, Wei-Chin Wu, Yu-Wen Chien, and Wen-Jin Cherng. Effect of aerobic interval

training on erythrocyte rheological and hemodynamic functions in heart failure patients with anemia. *International Journal of Cardiology*, 168(2):1243-1250, 2013.

7. Mathieu Gayda, Vincent Gremeaux, Joffrey Drigny, Martin Juneau, and Anil Nigam.

Muscle VO_2 and forearm blood flow repeatability during venous and arterial occlusion in healthy and coronary heart disease subjects. *Clinical hemorheology and microcirculation*, 59(2), 177–183, 2015

8. Terence E. Ryan, Jared T. Brizendine, and Kevin K. McCully. A comparison of exercise

type and intensity on the noninvasive assessment of skeletal muscle mitochondrial function using near-infrared spectroscopy. *Journal of Applied Physiology*, 114(2):230-237, 2013.

9. Andrew J. McNab and Lynn Stothers. Near-infrared spectroscopy: validation of

bladder-outlet obstruction assessment using noninvasive parameters. *Canadian Journal of Urology*. 15(5):4241-4248, 2008.

10. Zakary Grand Maison. Physics Analysis GUI (main). *GitHub*.

<https://github.com/zakgm2/PhysicsAnalysisGUI> , 2026

11. Zakary Grand Maison. PhysicsLibrary (Feat/-Data-Analysis). *GitHub*.

<https://github.com/zakgm2/PhysicsLibrary>, 2026

12. Mah, A. J., Nguyen, T., Ghazi Zadeh, L., Shadgan, A., Khaksari, K., Nourizadeh, M., Zaidi, A., Park, S., Gandjbakhche, A. H., & Shadgan, B. Optical Monitoring of Breathing Patterns and Tissue Oxygenation: A Potential Application in COVID-19 Screening and Monitoring. *Sensors (Basel, Switzerland)*, 22(19), 7274, 2022
13. Osada, T., Mortensen, S. P., & Rådegran, G. Mechanical compression during repeated sustained isometric muscle contractions and hyperemic recovery in healthy young males. *Journal of Physiological Anthropology*, 34, 36.
<https://doi.org/10.1186/s40101-015-0075-1>, 2015.